

Lesson 14 Proof by Induction

Prove the following by induction.

1. For every natural number n ,

a) $1 + 3 + 5 + \cdots + (2n - 1) = n^2$

b) $1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$

c) $1^2 - 2^2 + 3^2 - 4^2 + \cdots + (-1)^{n+1}n^2 = (-1)^{n+1} \frac{n(n+1)}{2}$

d) $1^3 + 2^3 + 3^3 + \cdots + n^3 = \frac{n^2(n+1)^2}{4}$

e) $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{n(n+1)} = 1 - \frac{1}{n+1}$

f) $n! \leq n^n$

g) (Geometric series) Let $r \in \mathbb{R}$,

$$1 + r + r^2 + \cdots + r^n = \frac{r^{n+1} - 1}{r - 1}$$

2. For every natural number n ,

a) $6|(n^3 - n)$

Hint: You may use the fact that product of two consecutive integers is even (Lesson 7 Proposition 5).

b) $3|(4^n - 1)$

3. Let F_j be the j th Fibonacci number. For every natural number n ,

a) $\sum_{j=1}^n F_j = F_{n+2} - 1$

b) $\sum_{j=1}^n F_{2j} = F_{2n+1} - 1$

4. Suppose $a_1 = 2$ and $a_n = 3a_{n-1} - 1$ for $n \geq 2$, then for all natural numbers n ,

$$a_n = \frac{3^n + 1}{2}$$

5. Suppose $a_1 = 5$ and $a_n = (a_{n-1})^2 - 15$ for $n \geq 2$, then a_n is a multiple of 5 for all $n \in \mathbb{N}$.

6. Consider the sequence,

$$s_1 = \sqrt{2}, \quad s_{n+1} = \sqrt{2 + s_n}$$

For all natural number n , $s_n < 2$.

7. Consider the sequence $a_1, a_2, a_3, \dots$ defined recursively as $a_1 = 1, a_2 = 1$, and $a_{n+1} = a_n + a_{n-1}$ for all integers $n \geq 2$. Use induction to prove that a_{3n} is even for all integers $n \geq 1$.

Hint: Determine a_3 and observe that for all integers $n \geq 1$,

$$a_{3n+3} = a_{3n+2} + a_{3n+1} \quad \text{and} \quad a_{3n+2} = a_{3n+1} + a_{3n}$$

The following exercises are optional.

8. Let $A_1, A_2, \dots, A_n$ be sets in a universal set. Then for $n \geq 2$,

a) $\overline{A_1 \cup A_2 \cup \dots \cup A_n} = \overline{A_1} \cap \overline{A_2} \cap \dots \cap \overline{A_n}$

b) $\overline{A_1 \cap A_2 \cap \dots \cap A_n} = \overline{A_1} \cup \overline{A_2} \cup \dots \cup \overline{A_n}$

You may assume De Morgan: $\overline{A_1 \cup A_2} = \overline{A_1} \cap \overline{A_2}$ and $\overline{A_1 \cap A_2} = \overline{A_1} \cup \overline{A_2}$.

9. Let p be a prime and $a_1, a_2, \dots, a_n$ be integers. If $n \geq 2$ and $p|a_1 \cdots a_n$, then $p|a_i$ for some $i = 1, \dots, n$.

Note: Recall that for any prime p , if $p|ab$, then $p|a$ or $p|b$.

10. A set with n elements has 2^n subsets, for any nonnegative integer n .

11. In the Tower of Hanoi puzzle, there are 3 poles. On the first pole, there is a stack of n disks arranged in increasing size with the largest at the bottom. The goal is to move the whole stack of disks to another pole, one disk at a time and at any time, no larger disk is on top of a smaller disk on any pole. Prove by induction that the puzzle can be solved in $2^n - 1$ moves.

Hint In the inductive step, break down the problem of moving a tower of $k + 1$ disks in Pole 1 to Pole 3 in 3 sub-problems as follows, and find the number of moves in each sub-problem:

- move the top k disks to Pole 2
- move the remaining disk in Pole 1 to Pole 3
- move the k disks in Pole 2 to Pole 3

